



CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

# Chapter 15: Similarity (With Applications to Maps and Models)

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

## 1 Similar triangles

1.  $\triangle ABC \sim \triangle EFG$  with  $\angle ABC = \angle EFG = 60^\circ$ ,  $AB = 15$  cm,  $BC = 3$  cm and  $EF = 75$  cm. Find  $FG$ .  
(a) 15 cm (b) 20 cm (c) 25 cm (d) 30 cm **ICSE 2025, 1 mark**

SOLUTION

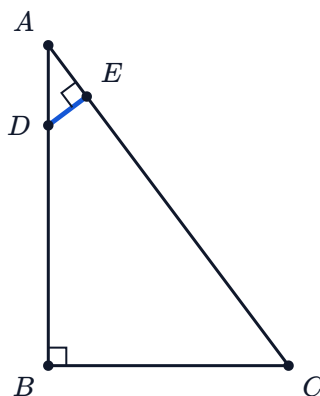
Corresponding sides are proportional:  $\frac{AB}{EF} = \frac{BC}{FG}$ .

$$\frac{15}{75} = \frac{3}{FG}, \text{ so } FG = 15 \text{ cm.}$$

**Answer: (a)**

## 2 Maps and models

2. In  $\triangle ABC$ ,  $\angle ABC = 90^\circ$ ,  $AB = 20$  cm and  $AC = 25$  cm.  $D$  is a point on  $AB$  and  $E$  is a point on  $AC$  with  $DE \perp AC$  and  $DE = 3$  cm. (a) Prove that  $\triangle ABC \sim \triangle AED$ . (b) Find  $BC$ ,  $AD$  and  $AE$ . (c)  $BCED$  is a plot of land drawn on a map, and its actual area is  $576 \text{ m}^2$ . Find the scale factor of the map. **ICSE 2025, 4 marks**



### SOLUTION

(a) In  $\triangle ABC$  and  $\triangle AED$ :  $\angle A$  is common and  $\angle ABC = \angle AED = 90^\circ$ . So  $\triangle ABC \sim \triangle AED$  (AA).

(b)  $BC = \sqrt{25^2 - 20^2} = 15$  cm. From the similar triangles,  $\frac{AB}{AE} = \frac{BC}{ED} = \frac{AC}{AD} = \frac{15}{3} = 5$ .  
 $AE = \frac{20}{5} = 4$  cm and  $AD = \frac{25}{5} = 5$  cm.

(c) Area of  $BCED$  on the map = area of  $\triangle ABC$  – area of  $\triangle AED = \frac{1}{2} \times 20 \times 15 - \frac{1}{2} \times 4 \times 3 = 150 - 6 = 144 \text{ cm}^2$ .

Actual area =  $576 \text{ m}^2 = 5760000 \text{ cm}^2$ .  $k^2 = \frac{144}{5760000} = \frac{1}{40000}$ , so  $k = \frac{1}{200}$ .

**Answer: (b)  $BC = 15$  cm,  $AD = 5$  cm,  $AE = 4$  cm (c) scale factor  $1 : 200$**

3. A model of a high-rise building is made to a scale of  $1 : 50$ . (a) The model is  $0.8$  m high; find the height of the building. (b) A flat in the building has a floor area of  $20 \text{ m}^2$ ; find its floor area in the model. **ICSE 2020, 3 marks**

### SOLUTION

(a) Height of building =  $0.8 \times 50 = 40$  m

(b) Areas are in the ratio  $1^2 : 50^2 = 1 : 2500$ . Area in the model =  $\frac{20}{2500} = 0.008 \text{ m}^2 = 80 \text{ cm}^2$ .

**Answer: (a)  $40$  m (b)  $0.008 \text{ m}^2$ , that is  $80 \text{ cm}^2$**

### 3 Conditions for similarity

4.  $\triangle ABC \sim \triangle PQR$ .  $AD$  bisects  $\angle BAC$  and meets  $BC$  at  $D$ ;  $PS$  bisects  $\angle QPR$  and meets  $QR$  at  $S$ . Which statement is true? (a)  $\triangle ABC \sim \triangle PQS$  (b)  $\triangle ABD \sim \triangle PQS$  (c)  $\triangle ABD \sim \triangle PSR$  (d)  $\triangle ABC \sim \triangle PSR$  **ICSE 2024, 1 mark**

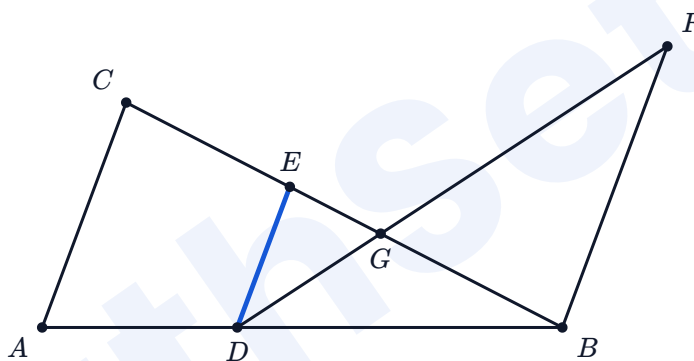
**SOLUTION**

$\angle B = \angle Q$  and  $\angle A = \angle P$  (similar triangles), so  $\angle BAD = \frac{1}{2}\angle A = \frac{1}{2}\angle P = \angle QPS$ .

Two angles of  $\triangle ABD$  equal two angles of  $\triangle PQS$ .

**Answer: (b)**

5.  $AC$ ,  $DE$  and  $BF$  are parallel, with  $D$  on  $AB$  and  $E$  on  $CB$ .  $DF$  meets  $CB$  at  $G$ . Given  $AC = 24$  cm,  $EG = 8$  cm,  $GB = 16$  cm and  $BF = 30$  cm, (a) prove that  $\triangle GED \sim \triangle GBF$  (b) find  $DE$  (c) find  $DB : AB$ . **ICSE 2023, 4 marks**



**SOLUTION**

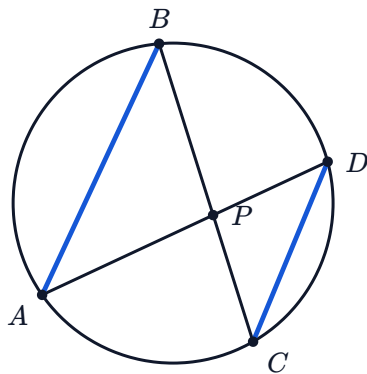
(a)  $\angle EGD = \angle BGF$  (vertically opposite) and  $\angle GED = \angle GBF$  (alternate angles,  $DE \parallel BF$ ). So  $\triangle GED \sim \triangle GBF$  (AA).

(b)  $\frac{DE}{BF} = \frac{EG}{GB}$ , so  $DE = 30 \times \frac{8}{16} = 15$  cm.

(c) In  $\triangle ABC$ ,  $DE \parallel AC$ , so  $\triangle BDE \sim \triangle BAC$  and  $\frac{DB}{AB} = \frac{DE}{AC} = \frac{15}{24} = \frac{5}{8}$ .

**Answer: (b)  $DE = 15$  cm (c)  $DB : AB = 5 : 8$**

6. Chords  $AD$  and  $BC$  of a circle intersect at  $P$ .  $AB = 9$  cm,  $PA = 7.5$  cm and  $PC = 5$  cm. (a) Prove that  $\triangle PAB \sim \triangle PCD$ . (b) Find  $CD$ . (c) Find area of  $\triangle PAB$  : area of  $\triangle PCD$ . **ICSE 2020, 3 marks**



**SOLUTION**

(a)  $\angle APB = \angle CPD$  (vertically opposite) and  $\angle PAB = \angle PCD$  (angles in the same segment, standing on the arc  $BD$ ). So  $\triangle PAB \sim \triangle PCD$  (AA).

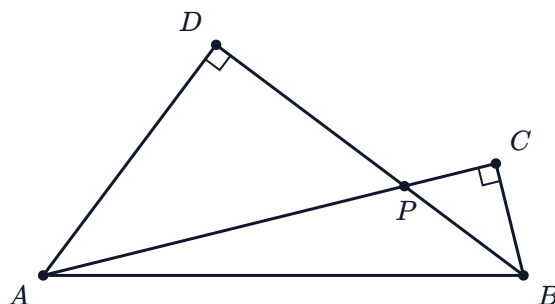
(b)  $\frac{AB}{CD} = \frac{PA}{PC}$ , so  $CD = \frac{9 \times 5}{7.5} = 6$  cm.

(c) Ratio of areas =  $\left(\frac{PA}{PC}\right)^2 = \left(\frac{7.5}{5}\right)^2 = \frac{9}{4}$

**Answer: (b)  $CD = 6$  cm (c)  $9 : 4$**

## 4 Areas of similar triangles

7.  $\triangle ADB$  and  $\triangle ACB$  are right-angled at  $D$  and  $C$ , on the same side of  $AB$ , and  $AC$  meets  $BD$  at  $P$ .  $AB = 10$  cm,  $AD = 6$  cm,  $BC = 2.4$  cm and  $DP = 4.5$  cm. (a) Prove that  $\triangle APD \sim \triangle BPC$ . (b) Find  $BD$  and  $PB$ . (c) Hence find  $PA$ . (d) Find area of  $\triangle APD$  : area of  $\triangle BPC$ . **ICSE 2024, 4 marks**



### SOLUTION

(a)  $\angle ADP = \angle BCP = 90^\circ$  and  $\angle APD = \angle BPC$  (vertically opposite). So  $\triangle APD \sim \triangle BPC$  (AA).

(b) In  $\triangle ADB$ :  $BD = \sqrt{10^2 - 6^2} = 8$  cm.  $PB = BD - DP = 8 - 4.5 = 3.5$  cm.

(c) From the similar triangles,  $\frac{PA}{PB} = \frac{AD}{BC} = \frac{6}{2.4} = 2.5$ , so  $PA = 2.5 \times 3.5 = 8.75$  cm.

(d) Ratio of areas =  $\left(\frac{AD}{BC}\right)^2 = (2.5)^2 = \frac{25}{4}$

**Answer: (b)  $BD = 8$  cm,  $PB = 3.5$  cm (c)  $PA = 8.75$  cm (d)  $25 : 4$**

Note: the measurements in this question do not fit together exactly (the Pythagoras theorem in  $\triangle ADP$  would give  $PA = 7.5$  cm). The word "hence" tells you to use the similar triangles, as done here.

## 5 Basic proportionality theorem

8. In  $\triangle PAB$ ,  $C$  is a point on  $PA$  and  $D$  is a point on  $PB$  with  $\angle BAP = \angle DCP = 70^\circ$ . If  $PC = 6$  cm and  $CA = 4$  cm, find  $PD : DB$ . (a)  $5 : 3$  (b)  $3 : 5$  (c)  $3 : 2$  (d)  $2 : 3$  **ICSE 2023, 1 mark**

### SOLUTION

The equal angles are corresponding angles, so  $CD \parallel AB$ .

By the basic proportionality theorem,  $\frac{PD}{DB} = \frac{PC}{CA} = \frac{6}{4} = \frac{3}{2}$ .

**Answer: (c)**

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